Q.30 Describe real-time clock (RTC) function with example. (CO3)

Q.31 Write a note on memory organisation of micro controller. (CO7)

Q.32 Explain SCADA system in details. (CO6)

Q.33 Explain interrupts and its types. (CO8)

Q.34 Explain serial port operation. (CO8)

Q.35 Explain LCD interface with microcontroller. (CO10)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Design a ladder diagram for a washing machine system. (CO4)

Q.37 Draw and explain memory organization in PLC with a diagram. (CO1)

Q.38 Write and explain instruction set of microcontroller. (CO8)

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Roll No. ....

5th Semester/ Elect
Subject : Programmable Logic Controllers & Micro Controllers

Time : 3 Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Ladder diagram is primarily used in : (CO1)

a) Structural design b) PLC programming

c) Flowchart d) Mechanical modelling

Q.2 A typical microcontroller has: (CO6)

a) RAM only

b) CPU and I/O only

c) ROM only

d) CPU, RAM, ROM and I/O ports

Q.3 DOL starter is used to: (CO5)

a) Increase voltage b) Stop motor

c) Start motor directly d) Control heating

Q.4 Function of ADD instruction is: (CO3)

a) Compare data b) Add data

c) Move data d) None

Q.5 A microcontroller differs from a microprocessor in: (CO5)

a) Only used in computers
b) Has built-in memory and I/O
c) Larger size
d) Higher power

Q.6 Which is NOT an arithmetic instruction? (CO3)

a) ADD b) SUB
c) MUL d) Mov

Q.7 MOV instruction copies (CO3)

a) Timers b) Counters
c) Data d) Programs

Q.8 Star-Delta starter is used in : (CO5)

a) Heating systems b) Industrial motors
c) Traffic lights d) Computers

Q.9 Which port is used for serial communication IN MCS-51? (CO7)

a) Port 0 b) Port 1
c) Port 2 d) Port 3

Q.10 Which component stores user program in PLC? (CO1)

a) RAM b) ROM
c) EEPROM d) Hard Disk

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

Q.11 State any two advantages of PLC. (CO1)

Q.12 Define retentive timer. (CO4)

Q.13 What is the role of sequencer instruction? (CO3)

Q.14 List types of comparison instructions. (CO3)

Q.15 Name two real-life applications of micro controllers (CO7)

Q.16 What is ladder logic used for? (CO4)

Q.17 Write a short note on power supply in PLC. (CO1)

Q.18 Mention features of SCADA system. (CO6)

Q.19 Write full form of SFR. (CO7)

Q.20 List any two application areas of SCADA. (CO6)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Describe the basic operation of PLC. (CO1)

Q.22 What is the purpose of counters in PLC? (CO3)

Q.23 List and explain Addressing Modes. (CO8)

Q.24 Explain the concept of self-holding relays. (CO1)

Q.25 How do timers function in a PLC? (CO3)

Q.26 Explain assembly language programming. (CO9)

Q.27 Write short notes on MOV and SUB instruction. (CO3)

Q.28 Explain different types of microcontroller and their applications. (CO11)

Q.29 Differentiate between Microcontroller and microprocessor. (CO7)